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Deep Learning and Its Applications

Dataset Proposal

Deep Learning on Large-Scale Data and Specialized Tasks

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1 Proposal Summary

This proposal selects the image-classification track for Assignment 2 and studies fine-grained food recognition using a reproducible ten-class subset of the Food-101 dataset. The project extends the end-to-end PyTorch pipeline developed in Assignment 1 to a larger RGB image dataset and compares a self-implemented convolutional neural network with an ImageNet-pretrained ResNet-50.

The ten classes are selected before experimentation as the first ten labels in alphabetical order: *apple pie*, *baby back ribs*, *baklava*, *beef carpaccio*, *beef tartare*, *beet salad*, *beignets*, *bibimbap*, *bread pudding*, and *breakfast burrito*. This rule is deterministic and prevents choosing classes after observing model performance.

Table 1: Assignment 2 proposal at a glance

Item	**Proposed choice**
Task track	Image classification
Dataset	Food-101, restricted to ten predefined classes
Data plan	7,500 official training images; 2,500 official test images; stratified 90/10 validation split with seed 42
Models	Custom CNN baseline and ImageNet-pretrained ResNet-50
Controlled factor	Frozen backbone versus full fine-tuning
Metrics	Accuracy and macro-F1
Approval status	Approved

2 Task and Motivation

2.1 Task Definition

The task is single-label image classification. Given an RGB image containing a pictured dish, the model predicts one label from the ten selected food categories. The input is an image tensor $x \in \mathbb{R}^{3\times224\times224}$ after preprocessing, and the output is a probability distribution over ten classes. The prediction unit is one image; each image contributes one ground-truth categorical label.

For a model f_θ with parameters θ, the predicted class is

$$\hat{y} = \underset{c\in\{1,\ldots,10\}}{\arg\max}\ f_\theta(x)_c.$$

Training will minimize cross-entropy loss over the training split. The main research question is:

> Does full fine-tuning of a pretrained visual backbone improve ten-class fine-grained food recognition enough to justify its additional training and inference cost compared with a frozen pretrained backbone and a custom CNN baseline?

This is a specialized task because visually similar dishes may differ in texture, ingredients, shape, and presentation, while the images may contain clutter, varied lighting, and inconsistent viewpoints. The training portion of Food-101 intentionally retains some label noise, making robustness and error analysis meaningful.

2.2 Assignment Scope

The group will work deeply on image classification only. The proposal covers dataset selection, task formulation, data splitting, leakage prevention, metrics, baseline design, pretrained-model design, fine-tuning, controlled experimentation, and compute planning. The instructor has approved this dataset scope and experimental plan. The implementation therefore proceeds under the recorded **Approved** decision.

3 Dataset Description

3.1 Source and Version

Food-101 was introduced by Bossard, Guillaumin, and Van Gool for food-category recognition [1]. The official ETH Zurich dataset page describes 101 food categories and 101,000 images. Each category contains 750 training images and 250 manually reviewed test images. The official source also states that training images were intentionally not cleaned and that all images were rescaled to a maximum side length of 512 pixels.

The official source and citation are:

- Dataset page: https://data.vision.ee.ethz.ch/cvl/datasets_extra/food-101/.
- Paper: Lukas Bossard, Matthieu Guillaumin, and Luc Van Gool, "Food-101 – Mining Discriminative Components with Random Forests," ECCV, 2014.
- Dataset version: the official Food-101 release accompanying the 2014 paper, downloaded from the ETH Zurich dataset page. The download URL, archive filename, and SHA-256 checksum will be recorded before implementation.
- License/usage: the official dataset page does not state a separate SPDX-style license. The group will use the archive only for this course and academic research, will not redistribute the images, and will preserve the usage terms and original image rights stated by the source.

The full dataset download is approximately 5 GB according to the official source. This proposal uses a deterministic subset of ten classes rather than changing the labels or creating new annotations.

3.2 Selected Classes and Sample Counts

The selected labels and planned counts are shown below. Since Food-101 provides 750 official training images and 250 official test images for each class, the subset contains 7,500 official training images and 2,500 official test images.

Table 2: Predefined ten-class subset

ID	Class	Train	Test
1	apple pie	750	250
2	baby back ribs	750	250
3	baklava	750	250
4	beef carpaccio	750	250
5	beef tartare	750	250
6	beet salad	750	250
7	beignets	750	250
8	bibimbap	750	250
9	bread pudding	750	250
10	breakfast burrito	750	250
Total		**7,500**	**2,500**

The selection exceeds Part III's requirements of at least five classes and at least 5,000 training samples. Every class has enough examples for stratified training, validation, and test evaluation. The final conclusions will be limited to this ten-class subset and will not be generalized to all 101 Food-101 categories.



Figure 1: Illustrative Food-101 image grid collected from the official dataset webpage; shown for visual context and not as a measured sample.

3.3 Annotations, Bias, and Limitations

Each image has one categorical food label. The dataset is balanced by the official class counts, but balance does not guarantee equal visual difficulty. Some categories may have similar appearance, and the training images may contain incorrect labels or distracting visual artifacts. The dataset may also underrepresent cuisines, preparation styles, and real deployment conditions outside the original collection.

The dataset is accessed from the official ETH Zurich source. Because that page does not provide a separate license label, this proposal does not claim that the dataset is freely redistributable or covered by a particular open-source license. Individual image rights remain subject to the terms of the original dataset and image sources.

4 Preliminary Distribution Analysis

The official distribution is class-balanced: each selected class contributes 750 training images and 250 manually reviewed test images. The first dataset inspection will verify these counts from the downloaded file structure and produce a class-count table.

The EDA will also inspect:

- image width, height, aspect ratio, and maximum dimension;
- unreadable files and invalid image channels;
- exact file hashes and perceptual near-duplicates;
- representative samples from every selected class;
- visually similar class pairs and likely ambiguity;
- the frequency of obvious label or image-quality problems in a documented sample.

The proposal records the known per-class distribution and the EDA procedure. Exact dimension, duplicate, and corruption counts will be generated from the downloaded archive rather than invented in advance.

5 Splitting and Leakage Prevention

The official Food-101 test split will remain untouched and will be reserved for one final evaluation after model and checkpoint selection. The 7,500 official training images will be divided at the image-record level using a stratified 90/10 split with random seed 42. This produces 6,750 training images and 750 validation images while preserving the same class proportions in both partitions. The fixed seed makes the manifest deterministic and reproducible across all model configurations:

Table 3: Planned data partitions

Partition	Images	Use
Training	6,750	Parameter fitting and augmentation
Validation	750	Model selection and checkpoint selection
Test	2,500	One-time final evaluation

The split unit is the individual image record. The split manifest will store the relative path, class label, split name, and seed. The group will check exact hashes before splitting and use a perceptual-duplicate check to identify near-identical images. Any excluded invalid file or duplicate will be recorded in a manifest with a reason.

No test image will be used for augmentation statistics, hyperparameter selection, early-stopping decisions, or checkpoint selection. All models and all controlled runs will use exactly the same manifests.

6 Preprocessing and Data Pipeline

Images will be converted to RGB and transformed to 224×224 tensors. Training augmentation will use random resized cropping, horizontal flipping, and moderate color augmentation. Validation and test images will use deterministic resizing and center cropping. All partitions will use ImageNet mean and standard-deviation normalization for compatibility with the pretrained model.

The planned data flow is shown in Figure 2. Training uses the augmentation branch in preprocessing, while validation and test use deterministic preprocessing.

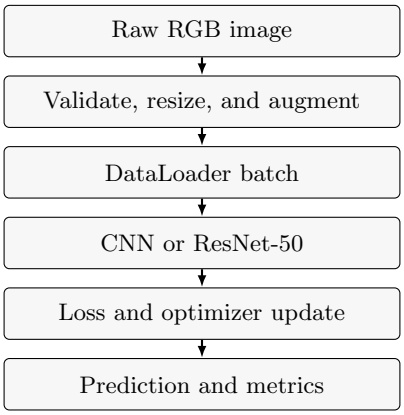


Figure 2: Planned Assignment 2 training and evaluation pipeline.

The diagram shows the forward path from the model through loss and optimizer update to prediction and metrics. During training, the model–loss–optimizer sequence repeats for the next batch; evaluation uses the prediction and metric stage without gradient updates.

The expected model input is a tensor of shape $(3, 224, 224)$, and a batch has shape $(B, 3, 224, 224)$. The preprocessing implementation will record one example batch and the final normalization values in the reproducibility notes.

7 Baseline and Pretrained Model Plan

7.1 Simple Baseline

The baseline will be a self-implemented CNN consisting of convolutional blocks, normalization, nonlinear activation, pooling, dropout, and a ten-class linear classification head. It will be trained from scratch with cross-entropy loss. The final report will record its architecture, parameter count, training time, inference latency, accuracy, and macro-F1.

The baseline is intentionally simple: it measures performance from task-specific learning without relying on transferred ImageNet features.

7.2 Pretrained Model

The main model will be an ImageNet-pretrained ResNet-50 [2] with its final classifier replaced by a ten-class head. The pretrained weights will use the official Torchvision model implementation and documentation. The fine-tuning procedure has two stages:

1. Freeze the ResNet backbone and train the new classifier head.
2. Unfreeze the full network and fine-tune with a lower learning rate for the pretrained layers.

The same data manifests, preprocessing policy, seed, evaluation metrics, and checkpoint selection rule will be used for both stages and for the custom CNN comparison.

8 Controlled Experiment and Evaluation

The mandatory controlled experiment will compare a frozen pretrained backbone with full fine-tuning under the same split manifests, seed, resolution, augmentation policy, optimizer family, evaluation metrics, and checkpoint-selection rule. The changed factor is whether the ResNet-50 backbone is trainable; the hypothesis is that full fine-tuning will improve macro-F1 while increasing training and inference cost.

Table 4: Controlled experiment design

Element	**Specification**
Hypothesis	Full fine-tuning improves macro-F1 on fine-grained food recognition.
Changed factor	ResNet-50 backbone frozen versus fully trainable.
Fixed factors	Split manifests, seed, resolution, augmentation, optimizer family, evaluation protocol, and checkpoint rule.
Metrics	Accuracy and macro-F1.
Decision rule	Prefer higher validation macro-F1; use test metrics and compute cost for final interpretation.
Configurations	Custom CNN, frozen-backbone ResNet-50, and fully fine-tuned ResNet-50.

Quantitative evaluation will report accuracy and macro-F1 because both are mandatory for Assignment 2 image classification. Secondary measurements will include parameter count, training duration, inference latency, and peak memory. Qualitative evaluation in the later report will show correct predictions, difficult correct predictions, and failures. The planned error taxonomy includes visual similarity, clutter or occlusion, lighting or framing problems, presentation variation, and suspected label noise.

For every reported experiment, the group will record the hypothesis, changed factor, fixed factors, metrics, decision criterion, hardware, software versions, seed, checkpoint rule, training time, and inference cost.

9 Compute Estimate and Reproducibility

The full Food-101 archive is approximately 5 GB before extraction. The ten-class subset requires less storage after filtering, but the original archive may be retained for traceability. The expected compute plan is:

Table 5: Planned compute budget

Item	**Proposal estimate or record to collect**
Resolution and batch	224×224 RGB; initial batch size 64 on a GPU with at least 8 GB memory
Training schedule	15 epochs for the custom CNN; 5 head-only plus 10 full-fine-tuning epochs for ResNet-50
Runs and time	Three main runs; approximately 2–4 GPU-hours total, with CPU expected to take substantially longer
Acceleration	CUDA and mixed precision if available; record hardware and software versions
Storage	Approximately 5 GB archive plus extracted subset, manifests, logs, and checkpoints
Inference	Measure warmed-up latency for a fixed batch of 64 images and report exact hardware

The proposal deliberately labels these as planning estimates. No training result, timing, accuracy, or hardware measurement is claimed before the experiments are actually run. Every final result will be traceable to a configuration file, split manifest, checkpoint, log or experiment identifier, and repository commit/tag.

9.1 Subset and Debugging Rules

The ten classes listed in Section 3.2 are fixed before experimentation and are used for all final comparisons. A smaller stratified subset may be used only to debug the data loader, loss computation, checkpointing, or evaluation code. Debugging results will not be reported as Assignment 2 results. The subset size, class rule, seed, and purpose will be recorded in the experiment configuration.

References

[1] Lukas Bossard, Matthieu Guillaumin, and Luc Van Gool, "Food-101 – Mining Discriminative Components with Random Forests," in *European Conference on Computer Vision*, 2014.

[2] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun, "Deep Residual Learning for Image Recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016.